

ABHINAV GOEL

Robotics Engineer | Autonomous Systems | UAV Engineering

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PROFESSIONAL SUMMARY

Robotics Engineer specializing in ROS2-based autonomous navigation, SLAM (Simultaneous Localization and Mapping), and vision-based UAV (Unmanned Aerial Vehicle) systems. Demonstrated track record of designing, simulating, and deploying autonomous mobile robots and drones from prototype to field-tested hardware. Proficient in sensor fusion (LiDAR, IMU, encoders), real-time computer vision (OpenCV, YOLO), drone autopilot integration (ArduPilot, MAVSDK, MAVLink), and embedded systems (ESP32, custom ESC design). Nationally ranked Top 16 out of 400+ teams in the e-Yantra Robotics Competition 2025–26 at IIT Bombay. Seeking an internship or research position in robotics, autonomous systems, or UAV engineering.

TECHNICAL SKILLS

Programming Languages:	Python, C++, Embedded C
Robotics Frameworks:	ROS2 Humble, Nav2 Navigation Stack, MoveIt2, SLAM Toolbox, AMCL, TF2, ROS2 Control
Drone & Autopilot:	ArduPilot, ArduPilot SITL, MAVLink, MAVSDK, DroneKit, PX4
Simulation Tools:	Gazebo Harmonic, Gazebo Ignition, RViz2
Computer Vision:	OpenCV, YOLOv4, YOLOv8, ArUco Marker Detection, Pixel-to-GPS Conversion, Visual Servoing
Robot Modeling & Control:	URDF, xacro, TF2, EKF (Extended Kalman Filter), PID Control, Sensor Fusion
Sensors & Hardware:	LiDAR, IMU (MPU6050), Ultrasonic Sensors, Barometer, Wheel Encoders, ESP32, ESP32-S3
Embedded Systems:	Arduino, Custom ESC Design (MOSFET-based), Motor Drivers, Embedded Firmware
Tools & Platforms:	Ubuntu Linux 22.04, Git, colcon, Arduino IDE, NumPy, Matplotlib, Pandas, Shapely, KML

EDUCATION

Bachelor of Technology (B.Tech) — Electronics and Communication Engineering

Aug 2024 – Aug 2028

PDPM Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Jabalpur

Relevant Coursework: Robotics (kinematics, Denavit–Hartenberg parameters, trajectory planning, mobile robot navigation, intelligent robots), Sensors and Actuators, Control Systems, Embedded Systems. Lab work included forward/inverse kinematics on a PUMA robot and a trajectory-following mobile robot (simulation and hardware).

EXPERIENCE

Coordinator — Electronics and Robotics Society

Aug 2024 – Present

PDPM IIITDM Jabalpur

- Organized robotics workshops, hands-on lab sessions, and technical events for 200+ student members.
- Mentored junior students on ROS2, embedded systems, and autonomous robot project development.
- Coordinated team participation in national competitions, including the e-Yantra Robotics Competition at IIT Bombay.

Core Committee Member — Aero Fabrication Club

Aug 2024 – Present

PDPM IIITDM Jabalpur

- Contributed to UAV design, fabrication, and flight testing activities within the club.
- Collaborated on drone hardware builds including frame assembly, ESC wiring, and flight controller setup.
- Participated in SAE Aerothon 2025 and NIDAR 2026 as part of the team UAV development effort.

PROJECTS

e-Yantra KrishiCobot — IIT Bombay (e-YRC 2025–26, Stage 2 Hardware Round)

Aug 2025 – Dec 2025

Tech: ROS2 Humble, LiDAR, Ultrasonic Sensors, IMU, Wheel Encoders, OpenCV, ArUco Markers, UR5 Robotic Arm, Visual Servoing

- Designed an autonomous agricultural robot for crop monitoring and manipulation with collision-free navigation using LiDAR, IMU, and encoder-based sensor fusion.
- Built a multi-sensor perception pipeline using LiDAR and ultrasonic sensors for obstacle detection and plant health assessment across all field lanes.
- Developed an ArUco marker-based localization system to identify fertilizer cans and classify healthy versus defective fruits using computer vision.
- Integrated a UR5 robotic arm with visual servoing for pick-and-place automation of fertilizer cans and defective fruits.
- Executed full experiments via remote hardware access to IIT Bombay equipment; achieved Top 16 ranking out of 400+ competing teams nationwide.

Disaster Surveillance Drone — NIDAR 2026

Jan 2026 – Jun 2026

Tech: Python, MAVSDK, ArduPilot SITL, OpenCV, YOLOv8, Shapely, KML, Matplotlib

- Built a polygon-based autonomous area survey mission covering approximately 75 acres in 20 minutes at 25 m altitude using optimized zigzag path planning.
- Integrated geofence protection with automatic Return-To-Launch (RTL) failsafe and real-time human detection using YOLOv8 with mission pause-center-resume logic.
- Automated the full mission lifecycle including upload, takeoff, survey execution, and RTL via the MAVSDK Python API.

Autonomous Drone Survey and Target Detection System — SAE Aerothon 2025

Jan 2025 – Jun 2025

Tech: Python, ROS2, DroneKit, ArduPilot SITL, MAVLink, Gazebo Harmonic, YOLOv4-Tiny

- Assembled a ROS2-based autonomous drone for survey missions and real-time object detection using YOLOv4-Tiny.
- Implemented pixel-to-GPS coordinate conversion using a pinhole camera model for precise target geolocation.
- Developed a PID-based centering control loop for precise target alignment during autonomous flight operations.

Differential Drive Robot — ROS2 Navigation and SLAM

Jan 2024 – Jun 2024

Tech: Python, ROS2 Humble, Gazebo Harmonic, Nav2, SLAM Toolbox, AMCL, URDF, xacro, TF2, EKF

- Modeled and simulated a four-wheeled differential drive robot with a complete URDF/xacro model and Gazebo physics environment.
- Achieved SLAM-based 2D occupancy map generation using SLAM Toolbox with AMCL pose tracking for robust localization.
- Deployed fully autonomous path planning, obstacle avoidance, and goal navigation using the Nav2 stack with EKF sensor fusion.
- Incorporated a gimbal-mounted camera with RViz2 visualization; produced a YouTube tutorial series covering the full pipeline (6 GitHub Stars).

ESP32-Based Quadcopter Drone — Elective Course Project

Jun 2024 – Dec 2024

Tech: ESP32-S3, MPU6050 IMU, Barometer, Coreless DC Motors, Custom MOSFET-based ESC

- Fabricated a quadcopter drone from scratch with IMU and barometer-based altitude sensing.
- Designed a custom MOSFET-based Electronic Speed Controller (ESC) and optimized motor thrust through gear reduction.
- Flashed manual flight control firmware on ESP32-S3 using Embedded C.

Mobile Robotic Platform — Course Project (Ongoing)

Aug 2024 – Present

Tech: ESP32, Motor Drivers, Sensors, Embedded C/C++, ROS2 (planned)

- Assembled an embedded mobile robotic platform with motor control and multi-sensor integration.
- Architected the ROS2-based autonomous navigation pipeline as the planned software layer.

ACHIEVEMENTS

- **e-Yantra Robotics Competition 2025–26 (IIT Bombay)** — Top 16 out of 400+ Teams: Qualified for the Stage 2 Hardware Round; ranked in the top 4% of participating teams nationwide.
- **SAE Aerothon 2025** — Selected to develop and present an autonomous drone survey and target detection system at the national level.
- **NIDAR 2026 — National UAV Competition** — Disaster Surveillance Drone project selected for participation in the national UAV challenge.